

Machine Learning (ML_PCOM7E) 2025

MSc Data Science
PGDip Data Science
MSc Artificial Intelligence
PGDip Artificial Intelligence
PGCert Artificial Intelligence

View Online



Allen Downey (2024) Think Python. Third edition. Sebastopol, CA: O'Reilly Media, Inc.
Available at:
https://learning.oreilly.com/library/view/think-python-3rd/9781098155421/?sso_link=yes&sso_link_from=university-of-essex.

Bishop, C.M. (2006a) Pattern Recognition and Machine Learning. Springer. Available at:
<https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/>.

Bishop, C.M. (2006b) Pattern Recognition and Machine Learning. Springer. Available at:
<https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/>.

Bishop, C.M. (2006c) Pattern Recognition and Machine Learning. Springer. Available at:
<https://www.microsoft.com/en-us/research/publication/pattern-recognition-machine-learning/>.

Brownlee, J. (2018) How to Calculate Correlation Between Variables in Python. Machine Learning Mastery. Available at:
<https://machinelearningmastery.com/how-to-use-correlation-to-understand-the-relationship-between-variables/>.

Calvello, M. (2024) Correlation vs. Regression: Key Differences and Similarities. G2. Available at: <https://www.g2.com/articles/correlation-vs-regression>.

Chatterjee, S. (2017) 'Good Data and Machine Learning'. Available at:
<https://medium.com/data-science/data-correlation-can-make-or-break-your-machine-learning-project-82ee11039cc9>.

Chernozhukov, Victor (2024a) 'Applied Causal Inference Powered by ML and AI'. Available at: <https://arxiv.org/abs/2403.02467>.

Chernozhukov, Victor (2024b) 'Applied Causal Inference Powered by ML and AI'. Available at: <https://arxiv.org/abs/2403.02467>.

Chernozhukov, Victor (2024c) 'Applied Causal Inference Powered by ML and AI'. Available at: <https://arxiv.org/abs/2403.02467>.

Chernozhukov, Victor (2024d) 'Applied Causal Inference Powered by ML and AI'. Available

at: <https://arxiv.org/abs/2403.02467>.

Crawford, S.L. (2006) 'Correlation and Regression', *Circulation*, 114(9). Available at: <https://doi.org/10.1161/CIRCULATIONAHA.105.586495>.

Criddle, C. (2020) Facebook sued over Cambridge Analytica data scandal. BBC. Available at: <https://www.bbc.co.uk/news/technology-54722362>.

DALL·E: Creating Images from Text (no date). Available at: <https://openai.com/blog/dall-e/>.

Diez-Olivan, A. et al. (2019) 'Data fusion and machine learning for industrial prognosis: Trends and perspectives towards Industry 4.0', *Information Fusion*, 50, pp. 92–111. Available at: <https://doi.org/10.1016/j.inffus.2018.10.005>.

Fuest, K and Krys, C (2018) Roland Berger Trend Compendium 2030 - Megatrend 5 Dynamic technology & innovation. Roland Berger. Available at: <https://www.rolandberger.com/en/Insights/Publications/Trend-Compendium-2030-Megatrend-5.html>.

Gary Metcalf, Susu Nousala, and David Ing (2024) *Industry 4.0 to Industry 5.0: Explorations in the Transition from a Techno-economic to a Socio-technical Future*. Singapore: Springer Nature Singapore. Available at: <https://link-springer-com.uniessexlib.idm.oclc.org/book/10.1007/978-981-99-9730-5>.

Goodfellow, I., Bengio, Y. and Courville, A. (2016a) *Deep learning*. Cambridge, Massachusetts: The MIT Press. Available at: <https://ebookcentral.proquest.com/lib/universityofessex-ebooks/detail.action?docID=6287197>.

Goodfellow, I., Bengio, Y. and Courville, A. (2016b) *Deep learning*. Cambridge, Massachusetts: The MIT Press. Available at: <https://ebookcentral.proquest.com/lib/universityofessex-ebooks/detail.action?docID=6287197>.

Goodfellow, I., Bengio, Y. and Courville, A. (2016c) *Deep learning*. Cambridge, Massachusetts: The MIT Press. Available at: <https://ebookcentral.proquest.com/lib/universityofessex-ebooks/detail.action?docID=6287197>.

Goyal, C. (2023) Intuition Behind Correlation - Definition and It's Types. *Analytics Vidhya*. Available at: <https://www.analyticsvidhya.com/blog/2021/04/intuition-behind-correlation-definition-and-its-types/>.

Harmadi, A.C. (2021) 10 Things to do when conducting your Exploratory Data Analysis (EDA). *Medium*. Available at: <https://medium.com/data-folks-indonesia/10-things-to-do-when-conducting-your-exploratory-data-analysis-eda-7e3b2dfbf812>.

Hutston, M. (2021) Robo-writers: the rise and risks of language-generating AI. *Nature*. Available at: <https://www.nature.com/articles/d41586-021-00530-0>.

Jordan, J. (2017) Evaluating a machine learning model. Available at: <https://www.jeremyjordan.me/evaluating-a-machine-learning-model/>.

Kugler, L. (2022) Artificial Intelligence, Machine Learning, and the Fight Against World Hunger. Communications of the ACM. Available at: <https://cacm.acm.org/magazines/2022/2/258220-artificial-intelligence-machine-learning-and-the-fight-against-world-hunger/fulltext>.

Leetaru, K. (2019) A Reminder That Machine Learning Is About Correlations Not Causation. Forbes. Available at: <https://www.forbes.com/sites/kalevleetaru/2019/01/15/a-reminder-that-machine-learning-is-about-correlations-not-causation/?sh=7b80959a6161>.

Machine Learning System Design Template | ML Interviews (no date). Available at: <https://www.machinelearninginterviews.com/ml-design-template/>.

'Machine Learning Tutorial Python - 8: Logistic Regression (Binary Classification)' (2018). YouTube: codebasics. Available at: <https://youtu.be/zM4VZR0px8E?feature=shared>.

Manuel Morales, Susu Nousala, and Morteza Ghobakhloo (2024) 'The Complexity of Sustainable Innovation, Transitional Impacts of Industry 4.0 to 5.0 for Our Societies: Circular Society Exploring the Systemic Nexus of Socioeconomic Transitions', in S. Nousala, G. Metcalf, and D. Ing (eds) Industry 4.0 to Industry 5.0. Singapore: Springer Nature Singapore, pp. 31-56. Available at: https://doi.org/10.1007/978-981-99-9730-5_2.

Mayo, M. (2017) Neural Network Foundations, Explained: Updating Weights with Gradient Descent & Backpropagation. KDnuggets. Available at: <https://www.kdnuggets.com/2017/10/neural-network-foundations-explained-gradient-descent.html>.

Miroslav Kubat (2021) An Introduction to Machine Learning. Cham: Springer International Publishing. Available at: <https://doi.org/10.1007/978-3-030-81935-4>.

Mostak, T. (no date) How Visualization is Transforming Exploratory Data Analysis. KDnuggets. Available at: <https://www.kdnuggets.com/2021/08/visualization-transforming-exploratory-data-analysis.html>.

Novaković, J.Dj. et al. (2017) 'Evaluation of Classification Models in Machine Learning', Theory and Applications of Mathematics & Computer Science, 7(1), pp. 39-46. Available at: <https://www.uav.ro/jour/index.php/tamcs/article/view/2234>.

Pant, A. (2019) Introduction to Machine Learning for Beginners. Medium. Available at: <https://medium.com/data-science/introduction-to-machine-learning-for-beginners-ee6024fdb08>.

Parisi, L. et al. (2022) 'Quantum ReLU activation for Convolutional Neural Networks to improve diagnosis of Parkinson's disease and COVID-19', Expert Systems with Applications, 187. Available at: <https://www.sciencedirect-com.uniessexlib.idm.oclc.org/science/article/pii/S0957417421012483>.

Patil, P. (2018) What is Exploratory Data Analysis? Medium. Available at: <https://medium.com/data-science/exploratory-data-analysis-8fc1cb20fd15>.

Pruciak, M. (2021) 10 Business Applications of Neural Network. Ideamotive. Available at: <https://www.ideamotive.co/blog/business-applications-of-neural-network>.

Schwab, K. (2016) The Fourth Industrial Revolution: what it means and how to respond. World Economic Forum. Available at: <https://www.weforum.org/agenda/2016/01/the-fourth-industrial-revolution-what-it-means-and-how-to-respond/>.

Sharma V, A. (2017) Understanding Activation Functions in Neural Networks. Medium. Available at: <https://medium.com/the-theory-of-everything/understanding-activation-functions-in-neural-networks-9491262884e0>.

sklearn.cluster.KMeans — scikit-learn 1.1.1 documentation (no date). Available at: <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>.

Snapshot Paper - AI and Personal Insurance - GOV.UK (no date). Available at: <https://www.gov.uk/government/publications/cdei-publishes-its-first-series-of-three-snapshot-papers-ethical-issues-in-ai/snapshot-paper-ai-and-personal-insurance>.

Snapshot Paper - Deepfakes and Audiovisual Disinformation - GOV.UK (no date). Available at: <https://www.gov.uk/government/publications/cdei-publishes-its-first-series-of-three-snapshot-papers-ethical-issues-in-ai/snapshot-paper-deepfakes-and-audiovisual-disinformation>.

Supervised learning (no date). scikit-learn. Available at: https://scikit-learn.org/stable/supervised_learning.html#supervised-learning.

This Person Does Not Exist - Random Face Generator (no date). Available at: <https://this-person-does-not-exist.com/en>.

Tippaya Thinsungnoen et al. (2015) 'The Clustering Validity with Silhouette and Sum of Squared Errors', in Proceedings of the 3rd International Conference on Industrial Application Engineering 2015. Available at: <https://www2.ia-engineers.org/conference/index.php/iciae/iciae2015/paper/view/576>.

Turck, M (2020) 'Data & AI Landscape 2020'. Available at: <https://www.mattturck.com/data2020>.

Valaskova, K. et al. (2018) 'Financial Risk Measurement and Prediction Modelling for Sustainable Development of Business Entities Using Regression Analysis', Sustainability, 10(7). Available at: <https://login.uniessexlib.idm.oclc.org/login?url=https://www.proquest.com/scholarly-journals/financial-risk-measurement-prediction-modelling/docview/2582920674/se-2?accountid=10766>.

Wall, M. (2019) Biased and wrong? Facial recognition tech in the dock. BBC. Available at: <https://www.bbc.co.uk/news/business-48842750>.

Walter, J. (2020) Deepfakes: The Dark Origins of Fake Videos and Their Potential to Wreak Havoc Online. Discover. Available at:
<https://www.discovermagazine.com/technology/deepfakes-the-dark-origins-of-fake-videos-and-their-potential-to-wreak-havoc>.

Wang, J et al. (2021) CNN Explainer: Learn Convolutional Neural Network. Available at:
<https://poloclub.github.io/cnn-explainer/>.

Xin Zhao (no date) 'Deep Learning in Social Computing', in Li Deng and Yang Liu (eds) Deep Learning in Natural Language Processing. Springer, Singapore, pp. 255–288.
Available at:
https://link-springer-com.uniessexlib.idm.oclc.org/chapter/10.1007/978-981-10-5209-5_9.